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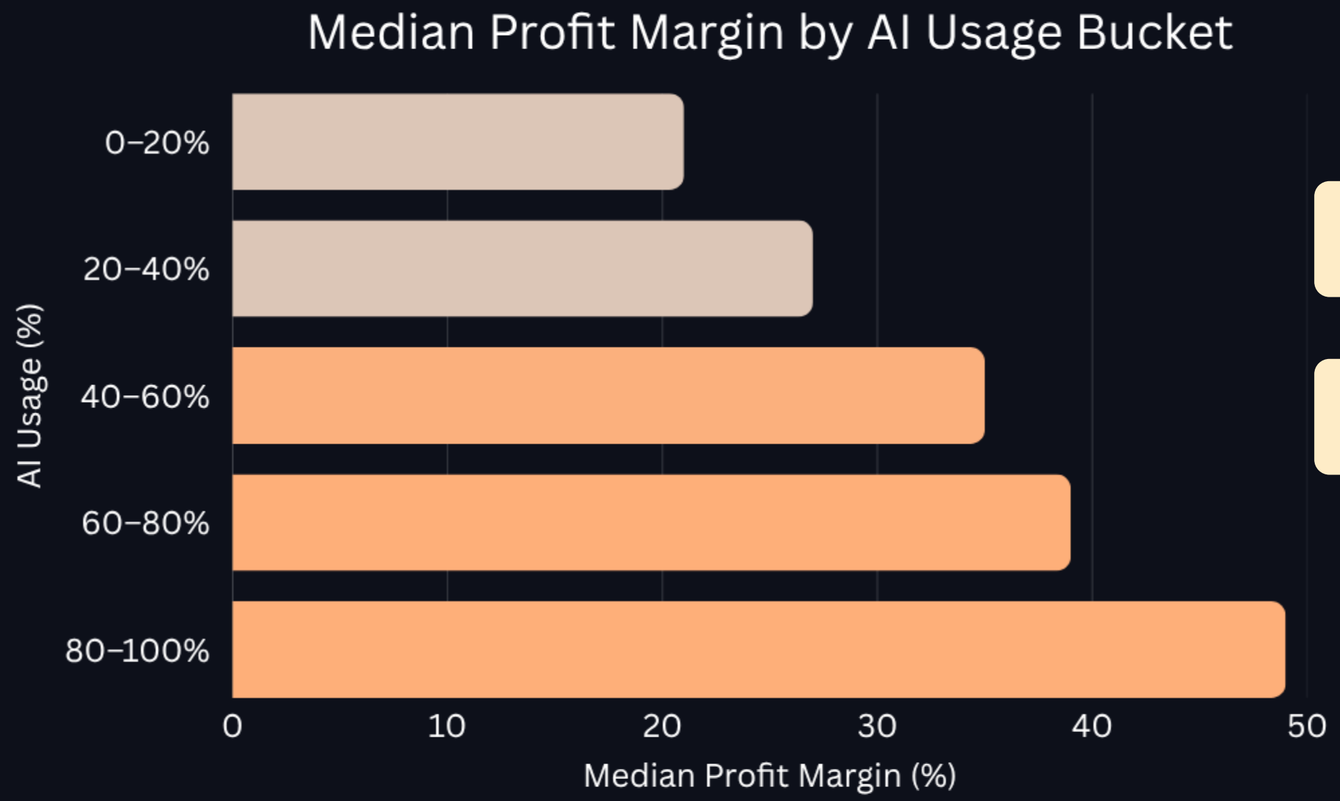
# AI PRODUCTIVITY ANALYSIS

## Beyond which AI usage threshold does rework destroy profit margin?

**AI MAKES WORK FASTER, BUT PROFIT DOES NOT AUTOMATICALLY FOLLOW**

| | | |
|---|---|---|
| Is there a threshold after which AI becomes costly? | Does value loss come from quality or revenue? | Who captures the efficiency gain? |

# AI-DRIVEN MARGIN GAINS ONLY EMERGE ABOVE 40% USAGE




**Below 40%: efficiency is absorbed by rework**

**Above 40%: gains translate into margin**

**+28pp margin increase (low → high AI)**

**Break point at 40–60% usage**

**45 % → 27 % SLA breach**

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# AI SAVES HOURS WHILE CONTRACTS DECIDE WHO CAPTURES VALUE.

Rework cost ≠ profit driver (Spearman ρ ≈ 0)
Speed gains dominate correction costs.



### HOW AI ACTS ON PROFIT

## PRODUCTIVITY GAINS ARE REAL.

Hours fall: 11.6 → 10.2

Rework rises: 0.133 → 0.298

### CONTRACT TYPE DECIDES OUTCOME

Fixed / Value-based → hours saved
=
margin captured

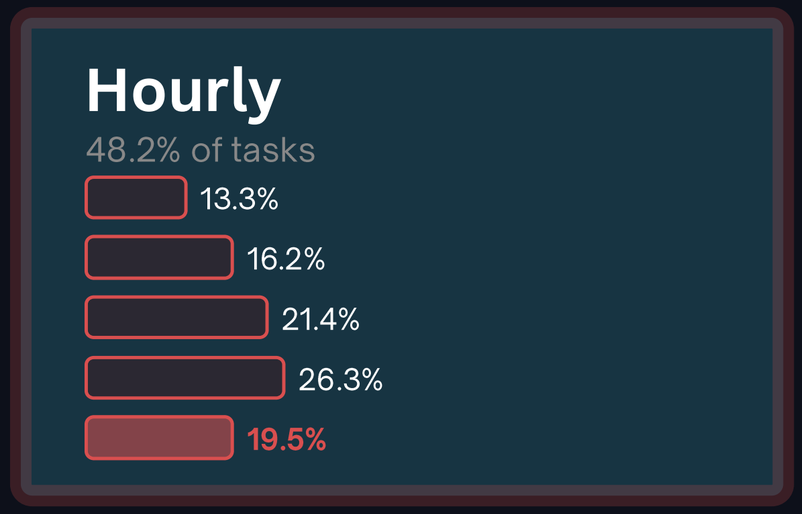
Hourly → hours saved = revenue lost
(26.3% → 19.5% at high AI)

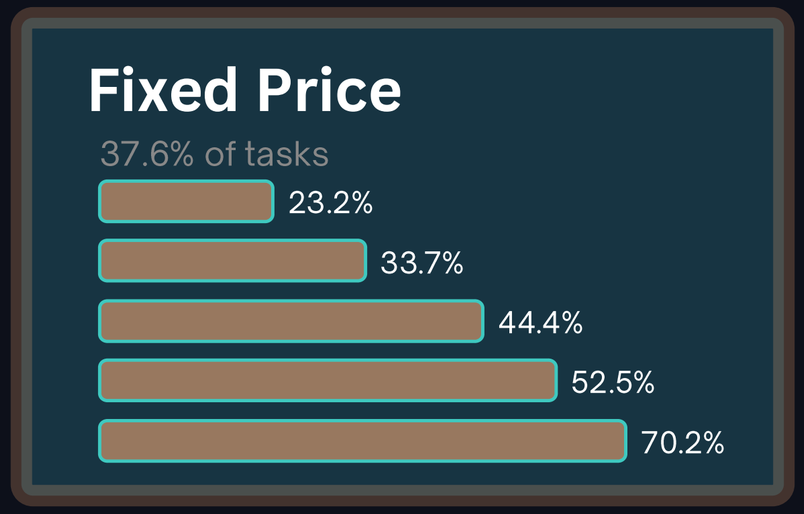
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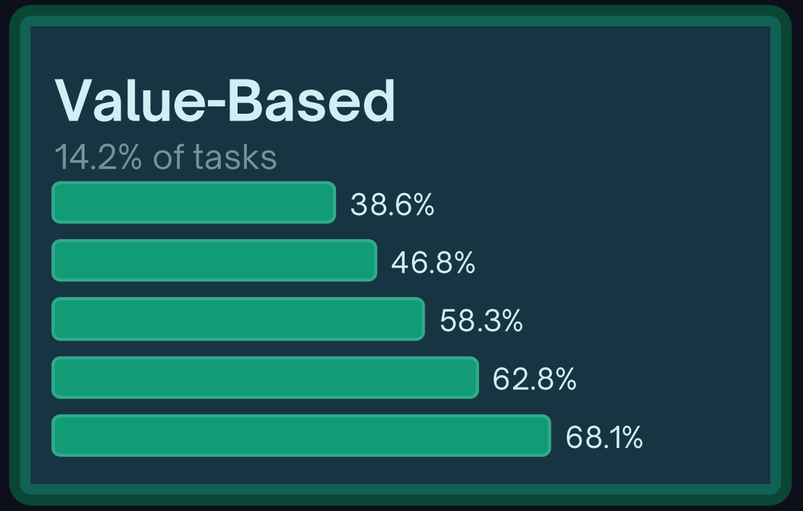
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# UNDER HOURLY CONTRACTS, AI CREATES VALUE FOR THE CLIENT — NOT FOR ALKEMY.

Hourly peaks at 26.3%, then drops to 19.5%
Fixed / Value-based continue rising (→ 70%+)



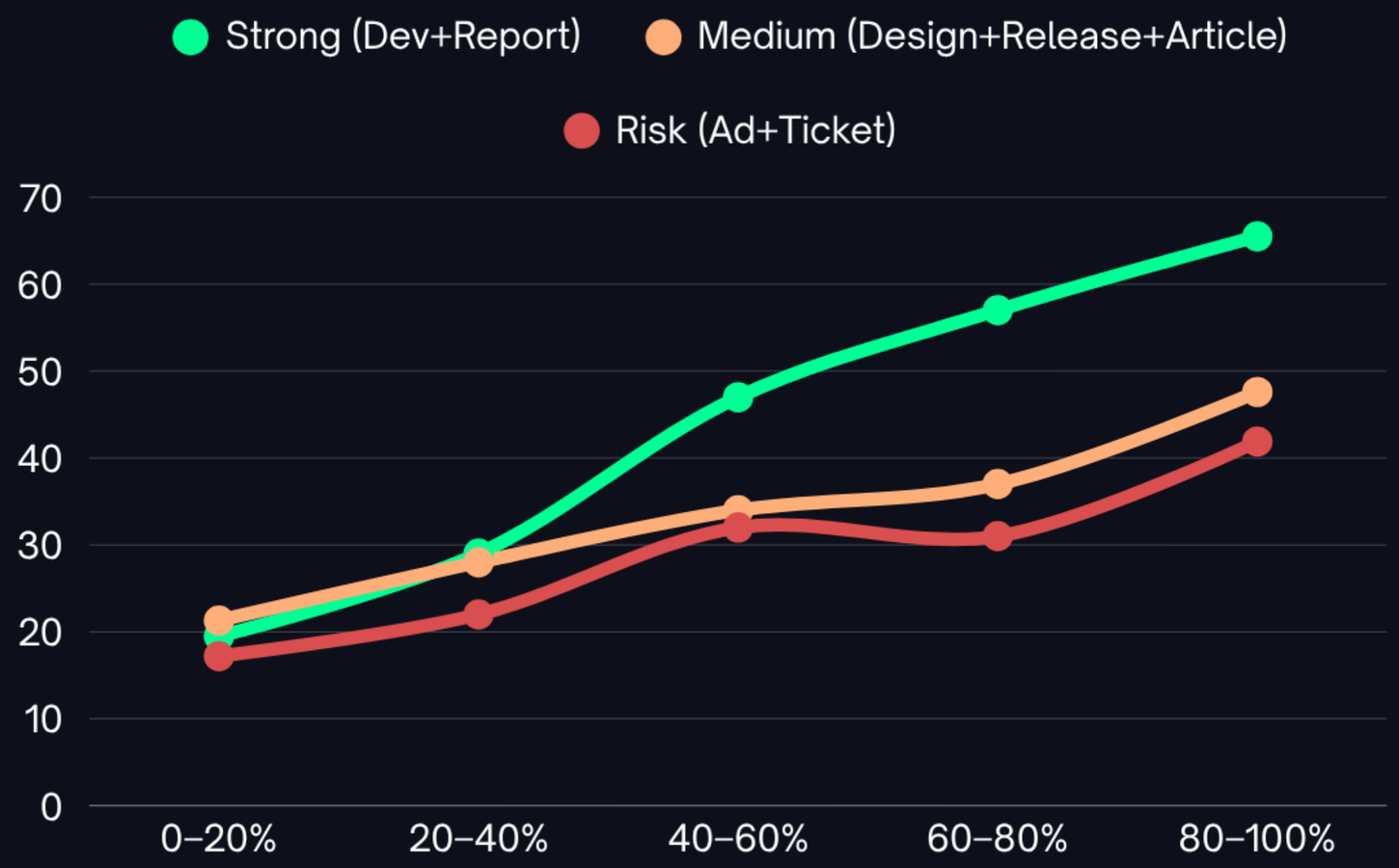



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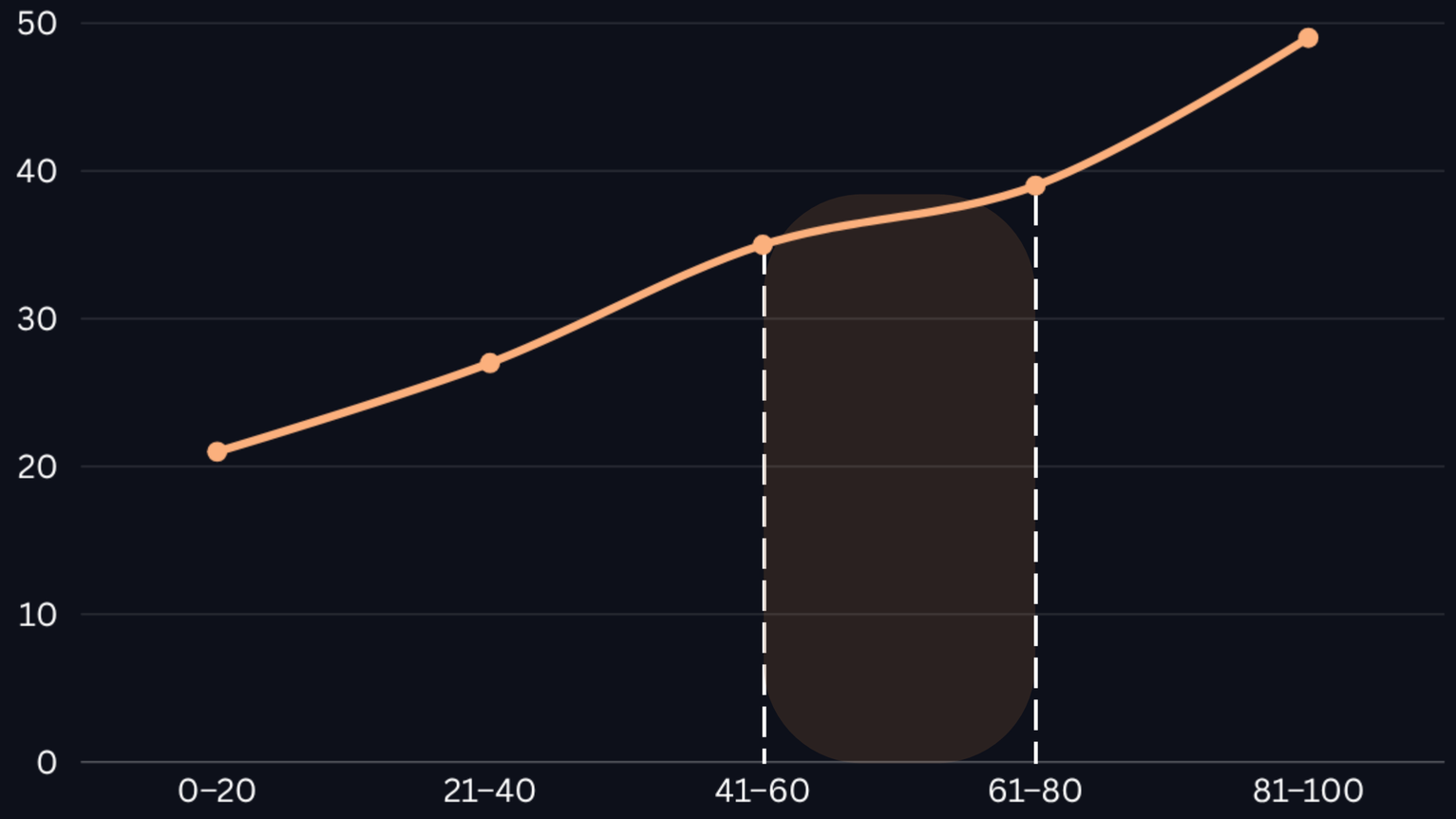
# HOW FAR YOU CAN PUSH AI DEPENDING ON THE TASK




**Only some tasks benefit from pushing AI further.**

# 40% THRESHOLD & 80% ACCELERATION

**The transition zone: where AI usage starts paying off**



# WHAT ACTUALLY DRIVES PROFIT (AND WHAT DOESN'T)

## PRICING MODEL

### THE STRUCTURAL DRIVER

Pricing model (hourly vs fixed/value-based) is the strongest driver of profit. It determines whether efficiency gains are captured or lost.

## BILLABLE RATIO

### THE EFFICIENCY SIGNAL

The share of worked time that becomes billable is a key predictor of profitability. It reflects operational efficiency and hidden slack.

## AI USAGE

### RANK 14

AI usage has a real but indirect effect. It does not independently drive profit — it amplifies the existing structure.

## MODELS AGREE ON THE SAME STORY

Linear Regression $R^2 \sim 0.20$

Random Forest $R^2 \sim 0.34$

**$\Delta R^2 = +0.14$ → strong non-linearity**

→ STRUCTURE DRIVES PROFIT. AI AMPLIFIES IT.

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# WHAT THE DATA REALLY SAYS

## 01

### AI is a multiplier

• does not move margin alone
• amplifies pricing
→ contract is the lever

## 02

### Binary view is wrong

• low AI ≠ better
• high AI → margin emerges
→ intensity matters

## 03

### Rework paradox

• rework ↑
• margin ↑
→ speed > quality cost

# WHAT AI MADE WRONG

• Suggested to remove outliers
• It would have removed real signal

→ reason why we kept them

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# WHAT THIS MEANS FOR DECISIONS

## 3 actions to capture AI value:

**Reprice Before Scaling AI**
hours → margin leak

**Set Thresholds**
40% = entry
60–80% = sweet spot

**Reduce Rework Strategically**
where it frees time
(not where it improves quality)

## What the data cannot capture (yet):

• CLIENT PRICING POWER • AI QUALITY DIFFERENCES • TRUE CAUSE OF REWORK

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# THANK YOU!